Homework 2 - Group

Panther Food Pantry Management System – Introduction and Project Domain

The Panther Food Pantry Management System is the proposed web-application domain for our project. The system is designed to provide a centralized platform for managing university food pantry services, including student assistance requests, appointments, food inventory, donations, and distributions. The project addresses the challenges that can occur when pantry activities depend on manual records, spreadsheets, and in-person communication. The main goal of the system is to improve the organization and efficiency of food pantry operations while providing students with a more convenient way to request and receive assistance. The system will maintain digital records of food inventory, student requests, appointments, donations, and distributions. This information can help pantry staff monitor available resources, process requests, and maintain accurate records.

The primary users of the Panther Food Pantry Management System are students, pantry staff, and administrators. Students will use the application to submit food assistance requests, schedule appointments, view request status, and review their distribution history. Pantry staff will manage requests, appointments, inventory, donations, and food distributions. Administrators will oversee users, monitor system activity, and access reports that support pantry management and decision-making. The overall workflow begins when a student submits a food assistance request or schedules an appointment. Pantry staff review and process the request before the student receives assistance. When food is distributed, staff record the transaction so that the system can maintain an updated inventory and distribution history. The basic workflow can therefore be represented as:

Student Request → Staff Review → Appointment → Food Distribution → Inventory Update → Record Completion

SDLC: Project Initiation and Planning

Our project begins with the project initiation and planning stage of the Software Development Life Cycle. During this stage, the team identifies the problem, defines the project scope, establishes objectives, identifies stakeholders and users, and determines the major functions that the application should support. The team will use the SDLC to organize the project from planning through deployment. The selected tools correspond to different stages of the development process. Jira will support project planning and management, Figma will support requirements analysis and interface design, Visual Studio Code will support implementation and debugging, GitHub and GitHub Actions will support version control and CI/CD, Postman will support testing and quality assurance, and Docker will support deployment and environment management.

Establishing the domain, scope, users, workflow, and responsibilities at the beginning provides a foundation for the team to move through the remaining SDLC stages in an organized manner.

The resulting development workflow is:

Initiation & Planning → Requirements & Design → Implementation → Version Control & CI/CD → Testing → Deployment → Evaluation

This structure allows the Panther Food Pantry Management System to demonstrate how the selected software tools can work together across the complete SDLC rather than functioning as separate technologies.

Jira - Project Planning and Management

Jira is a project management and issue-tracking tool designed to help software teams plan, organize, assign, and monitor development work. Its main features include Scrum and Kanban boards, task and issue tracking, backlogs,

sprint planning, workflow customization, priorities, deadlines, dashboards, and progress reporting. Jira's main advantages are its strong support for Agile development, detailed issue tracking, customizable workflows, and ability to manage complex software projects. However, its large number of features can create a learning curve, and configuration may be more complicated than simpler project-management tools.
Jira is widely used by software development and IT teams because it closely supports Agile practices and connects project planning with development activities. Asana, a major alternative, provides task lists, boards, timelines, deadlines, and team collaboration features. Asana is generally easier to learn and works well for broader business projects such as marketing campaigns, while Jira offers more specialized capabilities for software development, including bug tracking, sprint management, and technical workflows.
For the Panther Food Pantry Management System, Jira could organize work into requirements, development, testing, and documentation phases while assigning tasks to the eight team members and tracking progress.Individual Jira issues could represent features such as student requests, appointments, inventory management, donations, notifications, and reporting, allowing the team to track each feature from planning through testing and completion.

Figma - Requirements Analysis and Interface Design
Figma is a collaborative interface design and prototyping tool that helps software teams turn requirements into visual screen layouts and interactive prototypes. Its main features include wireframes, reusable components, shared design files, comments, and clickable prototypes. Its advantages include helping teams maintain consistent interfaces, gather feedback, and clarify how users will navigate the application before development begins. However, beginners may need time to learn its design tools, and the free Starter plan has limits on collaboration and advanced features.
Figma is used by design and development teams to coordinate interface design and developer handoff. For example, Workday uses Figma to improve collaboration between designers and developers. Penpot, an alternative, also supports collaborative interface design and prototyping and offers an open-source platform. Figma is a suitable choice for our project because its shared designs and prototypes would allow the team to review student and staff interactions before implementing them.
For the Panther Food Pantry Management System, Figma would translate the identified requirements into designs for student registration, login, food availability, assistance requests, appointment scheduling, and distribution history. Staff interfaces would include an administrative dashboard, inventory management, donation records, distribution tracking, and reports. A clickable prototype could demonstrate how a student requests assistance and schedules a pickup, allowing the team to identify missing steps or confusing navigation. After reviewing and refining the designs, developers would use them as a reference when building the web application.

Visual Studio Code

SDLC Stage: Implementation, coding, and debugging

Visual Studio Code (VS Code) is a free, cross-platform code editor developed by Microsoft. It supports the implementation phase of the software development life cycle by providing one workspace for writing, running, testing, and debugging code. Its core capabilities include syntax highlighting, intelligent code completion, code navigation, refactoring, an integrated terminal, debugging tools, testing support, and built-in Git source control. Additional languages and development tools can be added through extensions.

For the Panther Food Pantry Management System, VS Code would serve as the team's primary development environment. Developers could use it to build the student registration and appointment pages, administrative dashboard, inventory functions, and application programming interfaces. The integrated terminal would allow the team to install dependencies and run the application without leaving the editor. Its debugging tools could help identify errors in functions such as inventory updates, appointment scheduling, and request processing. The built-in

source-control interface would also allow developers to view changes, create branches, resolve conflicts, and commit code to GitHub.

VS Code's main advantages are that it is lightweight, customizable, available on Windows, macOS, and Linux, and supports many programming languages. Team members can install extensions for web development, code formatting, database access, Docker, and GitHub integration. The 2025 Stack Overflow Developer Survey identifies VS Code as one of the leading development environments, demonstrating its strong adoption within the software industry.

One limitation is that VS Code is not a complete development environment immediately after installation. Tools such as Git, programming-language runtimes, and many advanced capabilities must be installed separately or added through extensions. Installing too many extensions may also increase resource usage or cause compatibility problems.

Possible alternatives include IntelliJ IDEA and Eclipse for Java-focused development, WebStorm for specialized JavaScript and TypeScript development, and Visual Studio for large .NET and C++ applications. VS Code is recommended for this project because it is free, flexible, relatively easy to learn, and suitable for a collaborative web-development workflow.

Postman Testing and Quality Assurance

Purpose and core functionality. Postman is an API platform that supports the testing and quality assurance stage of the SDLC. Developers use it to send requests to an API, inspect the responses, and write tests that confirm each endpoint works correctly.

Main features. Its core features are saved request collections, environment variables, JavaScript test scripts, the Collection Runner, mock servers, and the Postman CLI for CI/CD pipelines.

Advantages. It is easy to learn, puts testing, mocking, and documentation in one tool, and runs the same tests manually or in a pipeline.

Limitations. The Free plan covers only one user, and team workspaces cost $19 per user per month, billed annually (Postman Pricing). Most features also require a Postman account, and it cannot test the browser interface.

Industry adoption. Postman reports more than 40 million developers and 500,000 organizations as users, including 98% of the Fortune 500 (Business Wire, 2026). Back-end developers and QA engineers use it to check endpoints and build regression test suites.

Comparison with alternatives. Insomnia and Bruno are open-source alternatives with local or Git storage, and Bruno needs no account at all. SoapUI is stronger for SOAP/XML services, while Swagger UI mainly provides interactive documentation rather than automated tests.

Integration into the Panther Food Pantry workflow. Each pantry endpoint, such as submitting a food request, gets saved requests and tests in a shared collection. GitHub Actions runs them with the Postman CLI on every pull request, and a failed run can be turned into a Jira issue (Postman Docs).

GitHub and GitHub Actions

SDLC Stage: Version control, collaboration, integration, and CI/CD

GitHub is a cloud-based platform that uses Git for version control, and teams make use of it in order to store their code, keep track of changes, work on different branches, and have the code reviewed via pull requests before it is merged. GitHub Actions includes CI/CD automation whereby workflows automatically build, test and deploy the application each time someone pushes code or opens a pull request.

The code for the Panther Food Pantry Management System would be kept on GitHub, with each feature having its own branch such as that for student requests, appointment scheduling, inventory, donations, and the admin dashboard. Before any changes were merged, team members would review the pull requests. GitHub Actions would carry out the unit tests, the code-quality checks, and the Postman API tests on every submission in order to detect any problems before they got to the main branch.

The main advantages of GitHub are its collaboration tools, the fact that it uses the widely adopted Git workflow, and its close integration with tools such as VS Code. Since the CI/CD processes take place within the repository, the team does not need to have a separate build server. A drawback is that GitHub Actions has limits on usage and storage, and complex workflow files can become difficult to maintain. Other options are GitLab and Bitbucket for hosting the code, Jenkins and CircleCI for CI/CD, and Azure DevOps, which combines both services. GitHub is the most suitable choice for this project since it provides the team with version control, collaboration, and automation all on a single platform.

Comparative Analysis

The proposed tools support different SDLC activities, so each should be evaluated against alternatives within its category. Our comparison considers functionality, configuration effort, collaboration, and suitability for the Panther Food Pantry Management System.

Selected Tool	Comparison and Trade offs	Application to the pantry project
Jira	Jira provides backlogs, Scrum and Kanban boards, and sprint planning. Asana supports broader project coordination through tasks and project views. We favor Jira for tracking software requirements and defects, although its workflows require more effort to configure	Organize student requests, appointments, and inventory features into assigned and prioritized work items.
Figma	Figma supports collaborative interface design and interactive prototypes. Penpot offers similar design and prototyping capabilities through an open-source platform. Figma fits the proposed design workflow; Penpot is a strong alternative when open-source tooling is a priority.	Review student request forms and staff dashboards before implementation.

Vs Code	VS Code provides an extensible editor with debugging support. WebStorm offers a dedicated JavaScript and TypeScript IDE. VS Code offers flexibility across languages, although the team must configure the required runtimes and extensions.	Develop and debug the application's interfaces, APIs, and inventory logic.
GitHub and GitHub Actions	GitHub Actions automates builds, tests, and deployment through repository-based workflows. GitLab provides comparable integrated CI/CD capabilities. Both require pipeline configuration and maintenance; GitHub Actions fits the team's proposed GitHub repository workflow.	Review code changes and automatically check them before integration.
Postman	Postman supports API request collections, scripted tests, and automated collection runs. Bruno provides a local, file-based approach suited to Git collaboration. Postman fits the proposed API-testing workflow, while Bruno is attractive when local collection storage is preferred.	Verify assistance requests, appointment validation, and inventory updates.
Docker	Docker packages applications and dependencies into containers. Podman offers a daemonless alternative for managing containers. Docker supports a consistent application environment, although the team must configure networking, persistent storage, and deployment infrastructure.	Run the application and its supporting services with repeatable configurations.

Recommended Toolset

We recommend Jira, Figma, VS Code, GitHub with GitHub Actions, Postman, and Docker for the Panther Food Pantry Management System. This combination covers the six selected SDLC categories and gives each tool a clear responsibility. Our recommendation is based on the project's need to coordinate eight contributors, review student and staff workflows, manage code changes, test application behavior, and maintain consistent environments.

For example, the team would first define a student appointment requirement in Jira and design its interface in Figma. Developers would implement it in VS Code and submit the changes through a GitHub pull request. A GitHub

Actions workflow would then run automated checks, including Postman API tests, before the changes are merged. Docker would package the application for consistent execution, and any defects found during testing would return to Jira for tracking and resolution. GitHub Actions supports automated build and test workflows (GitHub, n.d.), and Postman collections can run inside CI/CD pipelines through the Postman CLI (Postman, n.d.).

Contribution of Each Team Member

Team Member	**SDLC Stage**	**Contribution**
Usman Jibril	Project initiation and planning	Introduction and project domain, covering the project's goals, intended users, and overall workflow
Damario Johnson	Project planning and management	Jira section and compared it with Trello, Asana, Azure Boards, and Monday.com
Samuel	Requirements analysis and interface design	Figma section, compared it with Miro, Lucidchart, Penpot, and Sketch, and prepared the slide visuals
Gado	Implementation, coding, and debugging	Visual Studio Code section and compared it with IntelliJ IDEA, Eclipse, WebStorm, and Visual Studio
Oluwafemi Bamidele	Version control, collaboration, integration, and CI/CD	GitHub and GitHub Actions section and compared it with GitLab, Bitbucket, Jenkins, CircleCI, and Azure DevOps
Alvi Anowar	Testing and quality assurance	Postman section and compared it with Insomnia, Bruno, SoapUI, and Swagger UI
Franck Darole	Deployment, environment management, and maintenance	Researched and wrote the Docker section and compared it with Podman, containerd, Kubernetes, and virtual machines
Cecilio Kisitu	Evaluation across the complete SDLC	Comparative analysis and recommended toolset, organized the references

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